Exercise 3
Biasing and Current Sources

Winter Term 2025/26

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- For this exercise, the following parameters are to be used

$$\beta_{0n} = 360\ \mu A/V^2, \qquad V_{thn} = 0.193\ V, \qquad \lambda_n = 0.088\ V^{-1}$$
$$\beta_{0p} = 63.5\ \mu A/V^2, \qquad V_{thp} = 0.215\ V, \qquad \lambda_p = 0.023\ V^{-1}$$
$$L = 2\ \mu m$$

Exercise 3 – Task 1.a.1

1. Given are: $V_{\mathrm{DD}} = 1.2\ \mathrm{V},\ V_{\mathrm{GS1,2}} = 0.35\ \mathrm{V},\ I_{\mathrm{REF}} = 50\ \mu\mathrm{A},\ I_{\mathrm{OUT}} = 100\ \mu\mathrm{A}$

   – a.1. Calculate the widths of $\mathrm{T_R}$ and $\mathrm{T_1}$. Channel-length modulation is to be considered.

$$W_{\mathrm{T_R}} = \frac{2I_{\mathrm{REF}}L}{\beta_{0\mathrm{p}}\left(V_{\mathrm{DD}} - v_{\mathrm{GS1,2}} - V_{\mathrm{thp}}\right)^2\left[1 + \lambda_{\mathrm{p}}\left(V_{\mathrm{DD}} - v_{\mathrm{GS1,2}}\right)\right]}$$

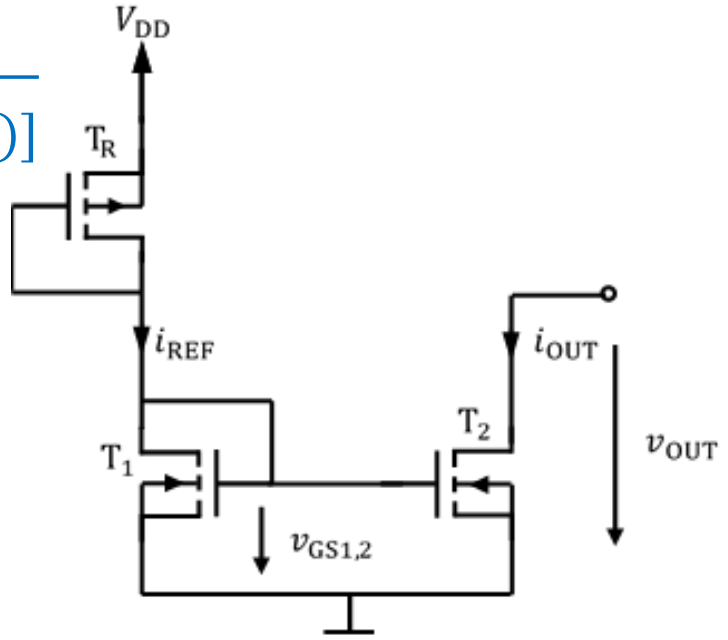
$$= \frac{2 \cdot 50\ \mu\mathrm{A} \cdot 2\ \mu\mathrm{m}}{63.5\ \frac{\mu\mathrm{A}}{\mathrm{V}^2}(1.2\ \mathrm{V} - 0.35\ \mathrm{V} - 0.215\ \mathrm{V})^2[1 + 0.023\ \mathrm{V}^{-1}(1.2\ \mathrm{V} - 0.35\ \mathrm{V})]}$$

$$= 7.7\ \mu\mathrm{m}$$

$$W_{\mathrm{T_1}} = \frac{2I_{\mathrm{REF}}L}{\beta_{0\mathrm{n}}\left(v_{\mathrm{GS1,2}} - V_{\mathrm{thn}}\right)^2\left[1 + \lambda_{\mathrm{n}}v_{\mathrm{GS1,2}}\right]}$$

$$= \frac{2 \cdot 50\ \mu\mathrm{A}\ \cdot 2\ \mu\mathrm{m}}{360\ \frac{\mu\mathrm{A}}{\mathrm{V}^2}(0.35\ \mathrm{V} - 0.193\ \mathrm{V})^2(1 + 0.088\ \mathrm{V}^{-1}0.35\ \mathrm{V})}$$

$$= 21.9\ \mu\mathrm{m}$$



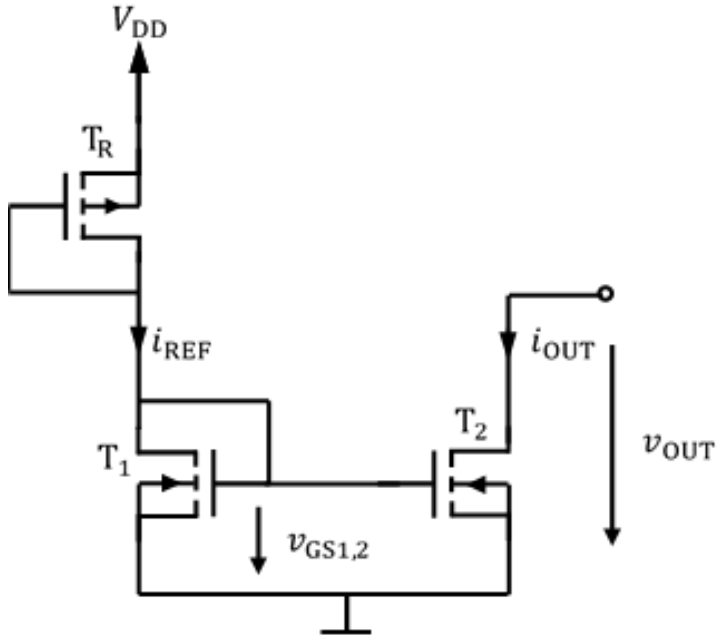
Exercise 3 – Task 1.a.2

– a.2. Determine the optimal value of v_{OUT} in order to precisely mirror i_{REF} two-to-one,

- i.e., $i_{\mathrm{OUT}} = 2i_{\mathrm{REF}}$. Calculate the width of T_2 accordingly.

$$v_{\mathrm{OUT,OPT}} = v_{\mathrm{GS1,2}}$$

$$W_{\mathrm{T}_2} = 2W_{\mathrm{T}_1} = 43.8\ \mu\mathrm{m}$$



Exercise 3 – Task 1.b)

b) Calculate the absolute and relative error of i_{OUT} for $V_{\mathrm{OUT}} = 1.2\ \mathrm{V}$!

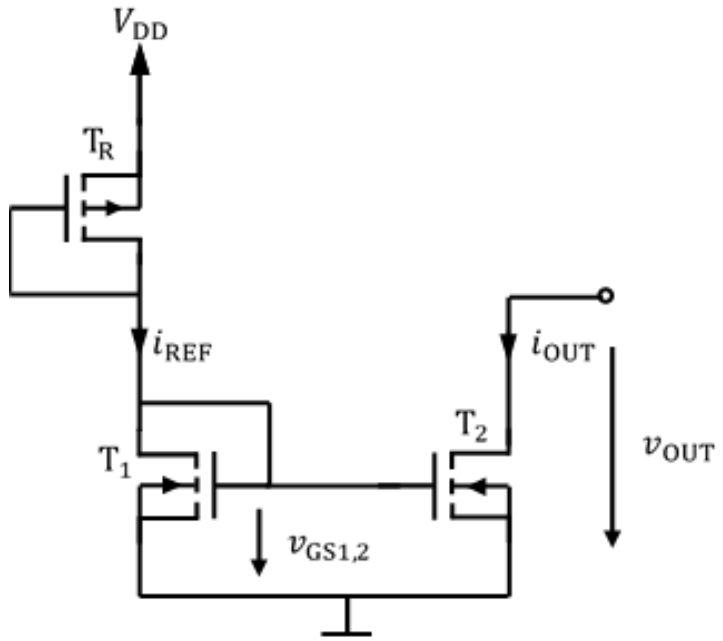
$$\frac{i_{\mathrm{OUT}}}{i_{\mathrm{REF}}} = \frac{\left(\frac{W}{L}\right)_2 \left[1 + \lambda_{\mathrm{n}}\left(v_{\mathrm{GS1,2}} + \Delta v_{\mathrm{DS}}\right)\right]}{\left(\frac{W}{L}\right)_1 \left[1 + \lambda_{\mathrm{n}} V_{\mathrm{GS1,2}}\right]}$$

$$= 2\left(1 + \frac{\lambda_{\mathrm{n}} \Delta v_{\mathrm{DS}}}{1 + \lambda_{\mathrm{n}} v_{\mathrm{GS1,2}}}\right)$$

$$= 2\left(1 + \frac{0.088\ \mathrm{V}^{-1} 0.85\ \mathrm{V}}{1 + 0.088\ \mathrm{V}^{-1} 0.35\ \mathrm{V}}\right)$$

$$\approx 2\ (1 + 0.725)$$

$$= 2.145$$

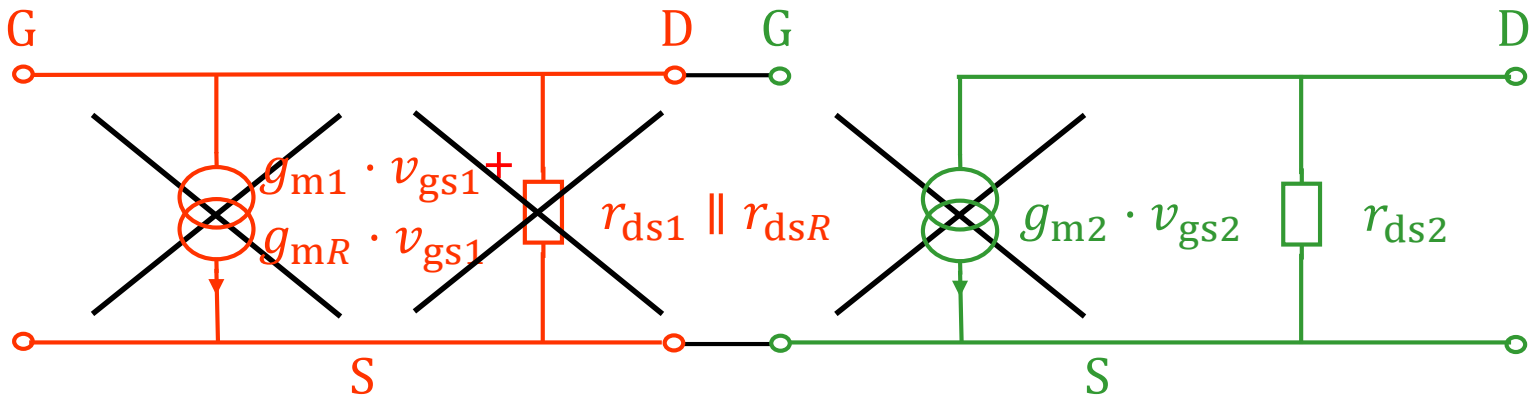


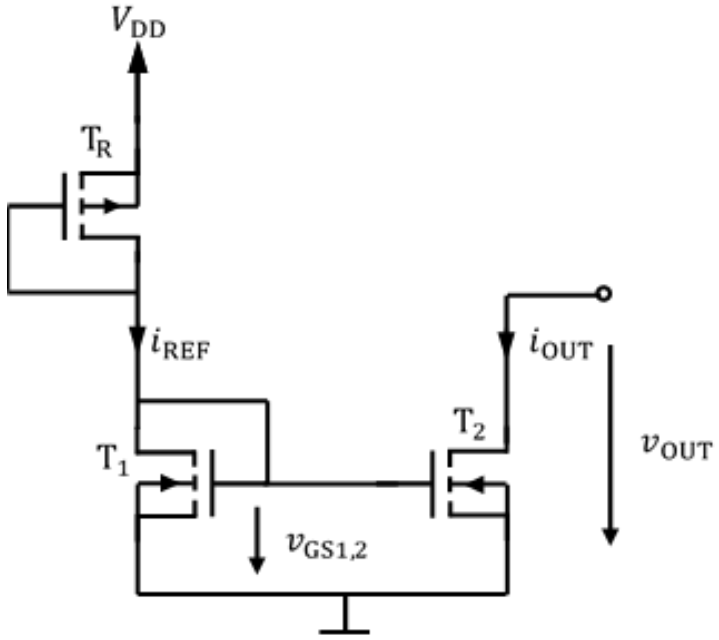
$$\rightarrow \Delta i = 7.25\%\ i_{\mathrm{OUT}} = \frac{7.25}{100} \times 100\mu\mathrm{A} = 7.25\mu\mathrm{A}$$

$$= 14.5\%\ i_{\mathrm{REF}}$$

Exercise 3 – Task 1.c)

c) Draw the small signal equivalent circuit of the current mirror. Either derive or state the output resistance and calculate its numerical value using the given parameters.





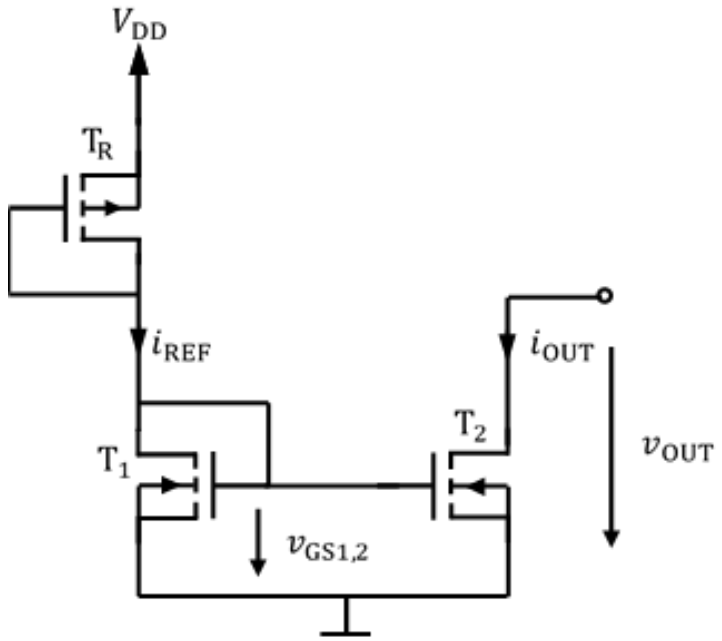
$$r_{out} = r_{ds2} = \frac{1}{\lambda_n I_{OUT}} = 114\ \text{k}\Omega$$

Exercise 3 – Task 1.d)

d) Determine the output swing of the current mirror!

$$V_{\mathrm{OUTmin}} = V_{\mathrm{GS1,2}} - V_{\mathrm{thn}} = 0.157\ \mathrm{V}$$
$$V_{\mathrm{OUTmax}} = V_{\mathrm{DD}} = 1.2\ \mathrm{V}$$

$$0.157\ \mathrm{V} < V_{\mathrm{OUT}} < 1.2\ \mathrm{V}$$



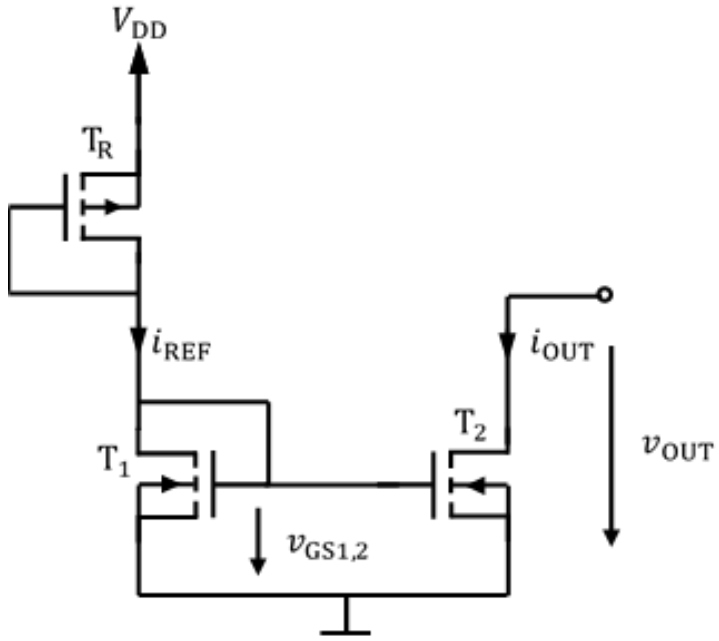
Exercise 3 – Task 1.e.1

e) Simulate your designed circuit using LTspice!

1. Connect a load resistor of $8.5\,\mathrm{k\Omega}$ between the output and V_{DD}. Perform a bias point simulation and check how well the voltages and currents match the given.

	specification	simulation
$V_{\mathrm{GS1,2}}$	0.35 V	0.354 V
V_{OUT}	0.35 V	0.333 V
I_{REF}	50 µA	50.36 µA
I_{OUT}	100 µA	101.89 µA

Good matching!

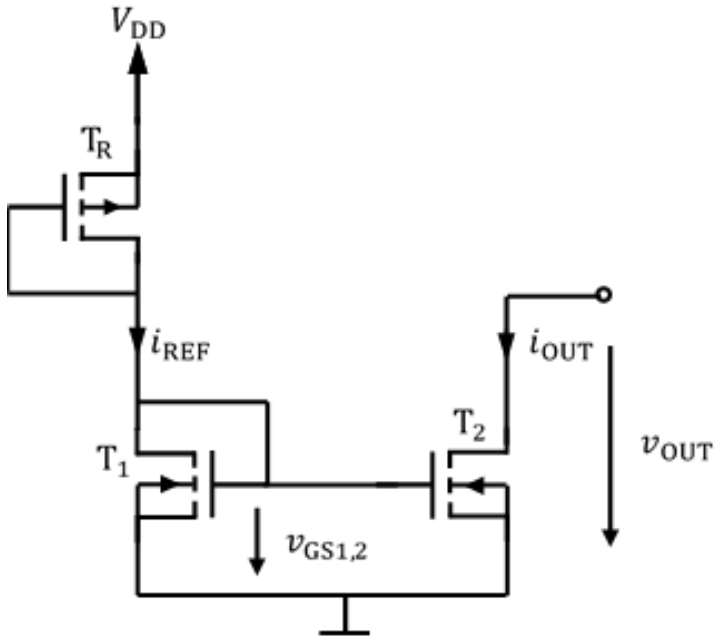


Exercise 3 – Task 1.e.2

e.2. Connect a load resistor of $8.5\,\mathrm{k\Omega}$ between the output and V_{DD}. Perform a bias point simulation and check how well the output resistance matches your calculation.

	calculation	simulation
r_{ds2}	$114\,\mathrm{k\Omega}$	$83.822\,\mathrm{k\Omega}$

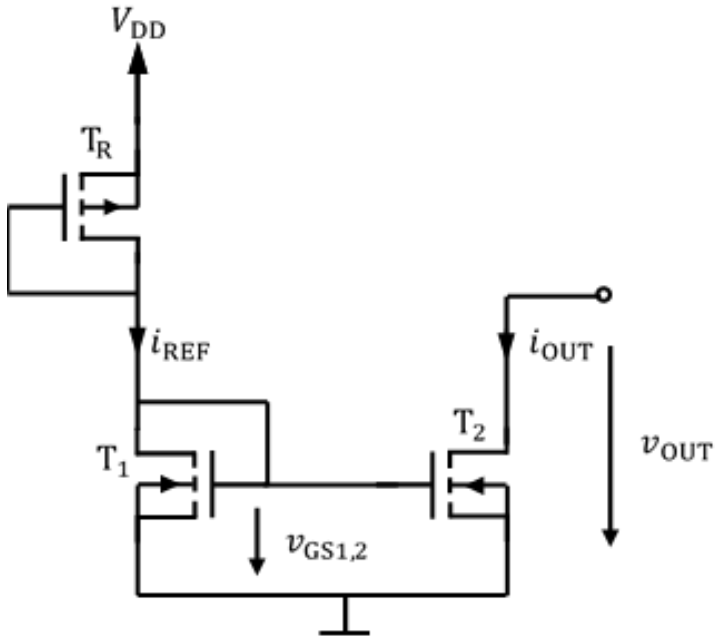
Transistor T2 is biased close to linear region!!



Exercise 3 – Task 1.e.3

- e.3. Perform a DC sweep for $V_{\mathrm{OUT}} = 0\ \mathrm{V} \ldots 1.2\ \mathrm{V}$ and check how well the output
  - swing matches your calculated values.

	calculation	simulation
V_{OUTmin}	$0.157\ \mathrm{V}$	$\approx\ 0.2\ \mathrm{V}$
V_{OUTmax}	$1.2\ \mathrm{V}$	$1.2\ \mathrm{V}$



Exercise 3 – Task 2.a)

2. Given are: $V_{\mathrm{DD}} = 1.2\ \mathrm{V},\ V_{\mathrm{GS1,2}} = 0.35\ \mathrm{V},\ I_{\mathrm{REF}} = 50\ \mu\mathrm{A},\ I_{23} = I_{\mathrm{OUT}} = 100\ \mu\mathrm{A}$

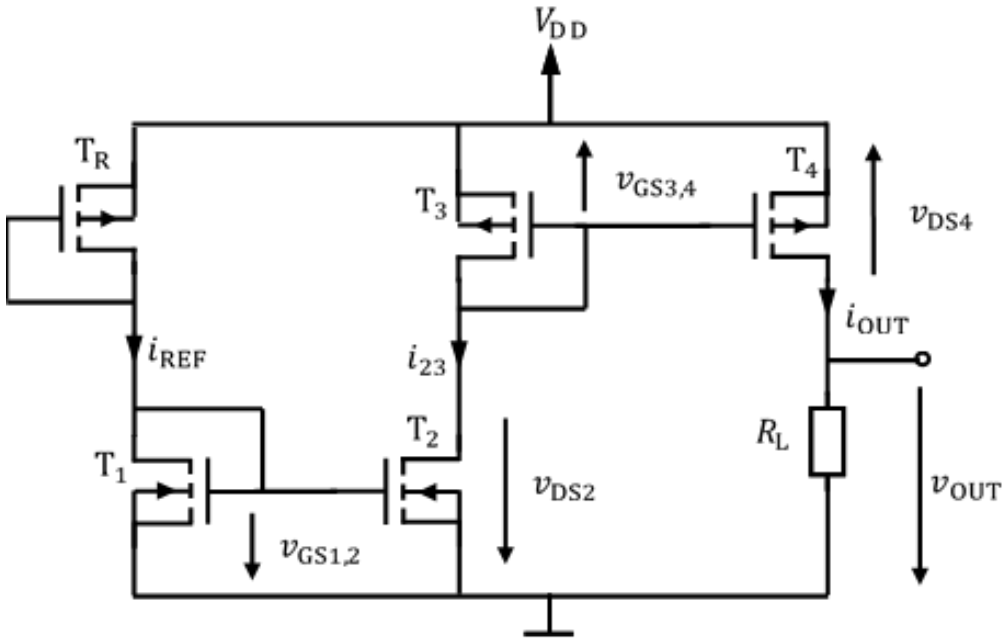
   a) Determine the optimal value of v_{DS2} in order to precisely mirror i_{REF} tow-to-one, i.e., $i_{23} = 2i_{\mathrm{REF}}$. Calculate the width of T_3 accordingly.

$$v_{\mathrm{DS2,OPT}} = v_{\mathrm{GS1,2}} = 0.35\ \mathrm{V}$$

$$W_{\mathrm{T}_3} = \frac{2I_{23}L}{\beta_{0\mathrm{p}}\left(V_{\mathrm{DD}} - v_{\mathrm{DS2,OPT}} - V_{\mathrm{thp}}\right)^2\left[1 + \lambda_{\mathrm{p}}\left(V_{\mathrm{DD}} - v_{\mathrm{DS2,OPT}}\right)\right]}$$

$$= \frac{2 \cdot 100\ \mu\mathrm{A} \cdot 2\ \mu\mathrm{m}}{63.5\ \frac{\mu\mathrm{A}}{\mathrm{V}^2}(1.2\ \mathrm{V} - 0.35\ \mathrm{V} - 0.215\ \mathrm{V})^2 \cdot [1 + 0.023\ \mathrm{V}^{-1}(1.2\ \mathrm{V} - 0.35\ \mathrm{V})]^{-1}}$$

$$= 15.4\ \mu\mathrm{m}$$



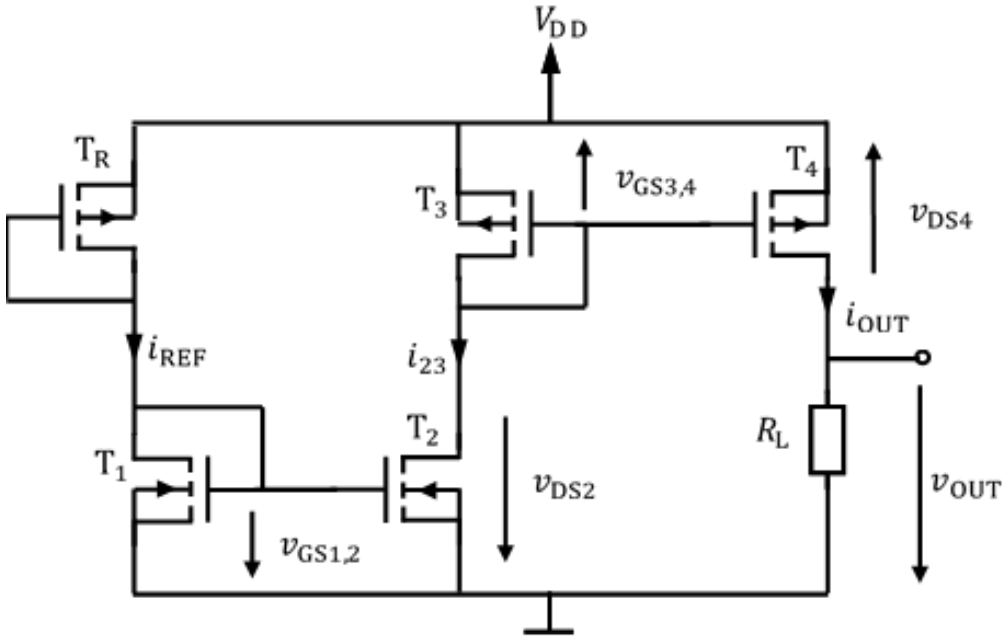
Exercise 3 – Task 2.b)

b) Determine the optimal value of v_{DS4} in order to precisely mirror i_{23} one-to-one, i.e., $i_{\mathrm{OUT}} = i_{23} = 2i_{\mathrm{REF}}$. Calculate the width of T_4 and the resistance R_{L} accordingly.

$$v_{\mathrm{DS4opt}} = v_{\mathrm{DS3opt}} = V_{\mathrm{DD}} - v_{\mathrm{GS1,2}} = 0.85\ \mathrm{V}$$

$$R_L = \frac{V_{\mathrm{DD}} - v_{\mathrm{DS4opt}}}{i_{\mathrm{OUT}}} = \frac{1.2\ \mathrm{V} - 0.85\ \mathrm{V}}{100\ \mu\mathrm{A}} = 3.5\ \mathrm{k\Omega}$$

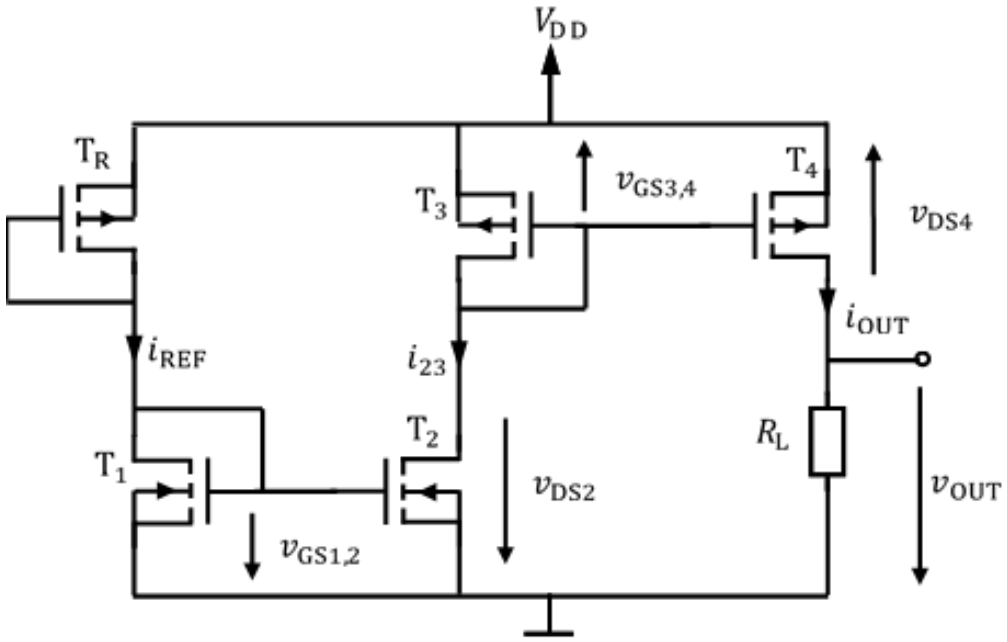
$$W_{\mathrm{T}_4} = W_{\mathrm{T}_3} = 15.3\ \mu\mathrm{m} \text{ since } i_{\mathrm{OUT}} = i_{23}$$



Exercise 3 – Task 2.c)

c) Calculate the output resistance of the current mirror neglecting $\mathrm{R_L}$.

$$r_{\mathrm{out}} = r_{\mathrm{ds4}} = \frac{1}{\lambda_{\mathrm{p}} I_{\mathrm{OUT}}} = \frac{1}{0.023\ \mathrm{V}^{-1} 100\ \mu\mathrm{A}} = 435\ \mathrm{k\Omega}$$

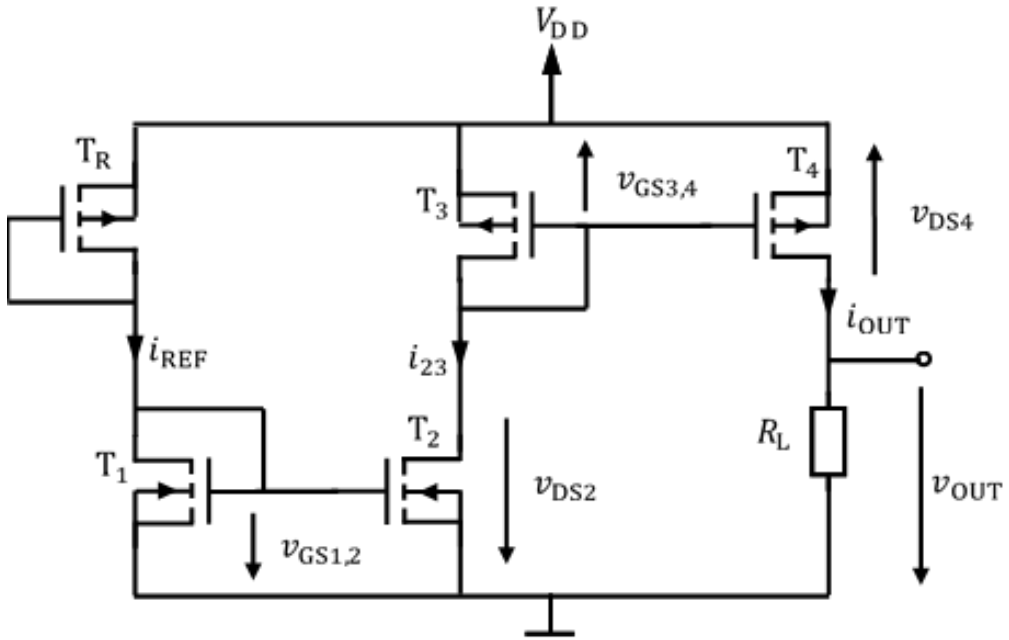


Exercise 3 – Task 2.d)

d) Calculate the resistance R_L for which T_4 operates in saturation.

$$
\begin{aligned}
v_{\mathrm{OUTmax}} &= V_{\mathrm{DD}} - v_{\mathrm{DS4min}} \\
&= V_{\mathrm{DD}} - \left(v_{\mathrm{GS3,4}} - V_{\mathrm{th,P}}\right) \\
&= 1.2\ \mathrm{V} - (0.85\ \mathrm{V} - 0.215\ \mathrm{V}) \\
&= 0.565\ \mathrm{V}
\end{aligned}
$$

$$
\Rightarrow R_{\mathrm{Lmax}} = \frac{v_{\mathrm{OUTmax}}}{I_{\mathrm{OUT}}} = \frac{0.565\ \mathrm{V}}{100\ \mu\mathrm{A}} = 5.65\ \mathrm{k}\Omega
$$

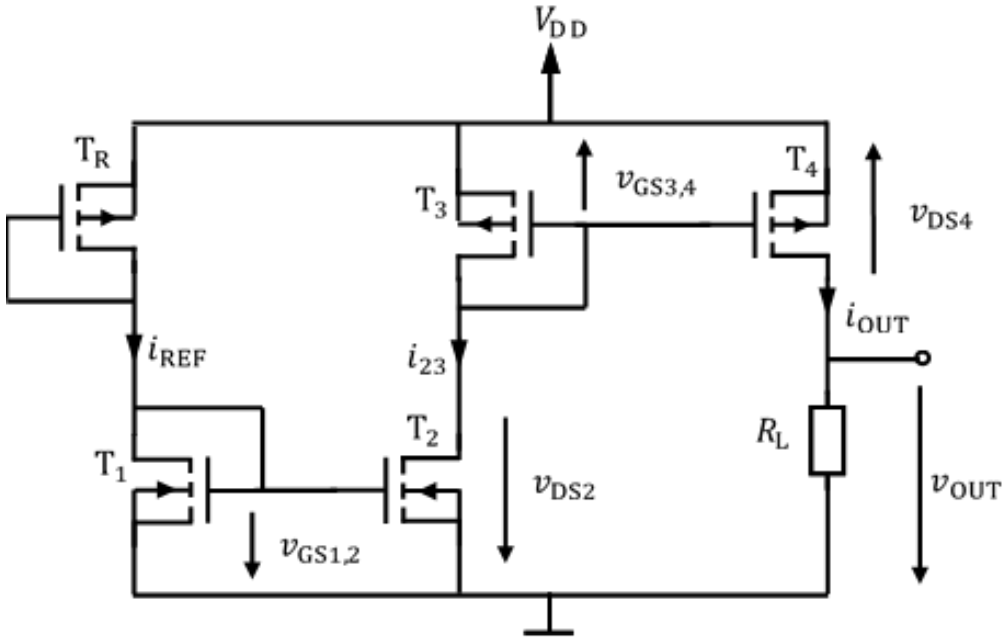


Exercise 3 – Task 2.e.1

e) Simulate your designed circuit using LTspice.

1. Perform a bias point simulation and check how well the voltages and currents match the given values.

	specification	simulation
$V_{GS1,2}$	0.35 V	0.3540 V
V_{DS2}	0.35 V	0.3504 V
$V_{GS3,4}$	0.85 V	0.8495 V
V_{DS4}	0.85 V	0.8428 V
V_{OUT}	0.35 V	0.3572 V
I_{REF}	50 µA	50.36 µA
I_{23}	100 µA	102.09 µA
I_{OUT}	100 µA	102.08 µA

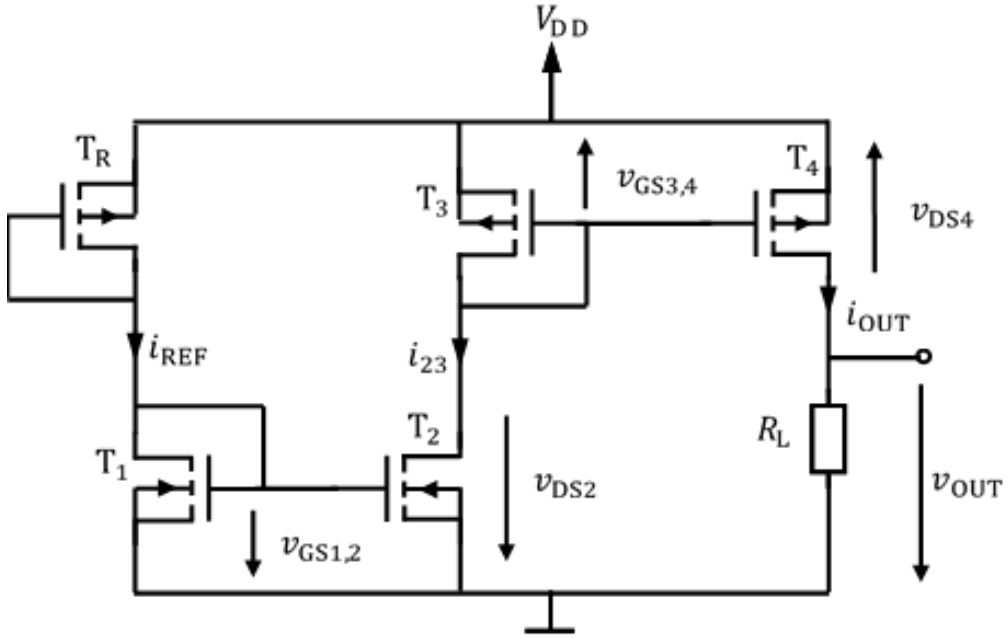


Exercise 3 – Task 1.e.2

- e.2. Perform a bias point simulation and check how well the output resistance
  - matches your calculation.

	calculation	simulation
r_{ds4}	$435\ \mathrm{k\Omega}$	$454.55\ \mathrm{k\Omega}$

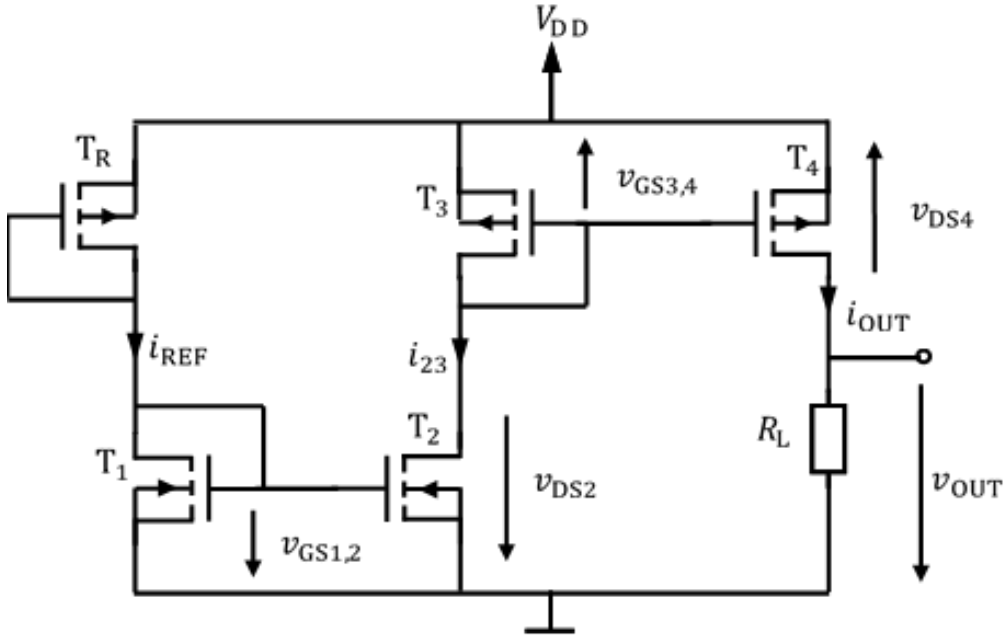
Biased in deep saturation



Exercise 3 – Task 1.e.3

e.3. Perform a parametric sweep for $R_\mathrm{L} = 100\ \Omega \ldots 100\ \mathrm{k}\Omega$ and check how well the maximum R_L matches your calculation.

	calculation	simulation
R_Lmax	$5.65\ \mathrm{k}\Omega$	$\approx 5.8\ \mathrm{k}\Omega$



Exercise 3 – Task 1.e.4

- e.4. Perform a DC sweep for $v_{\mathrm{OUT}} = 0\ \mathrm{V} \ldots 1.2\ \mathrm{V}$ ($\mathrm{R_L}$ disconnected) and check how
  - well the output swing matches your calculation.

	calculation	simulation
v_{OUTmin}	$0\ \mathrm{V}$	$0\ \mathrm{V}$
v_{OUTmax}	$0.565\ \mathrm{V}$	$\approx\ 0.62\ \mathrm{V}$

Good matching!

